New ADC drug design strategy based on the difference of glutamine metabolism

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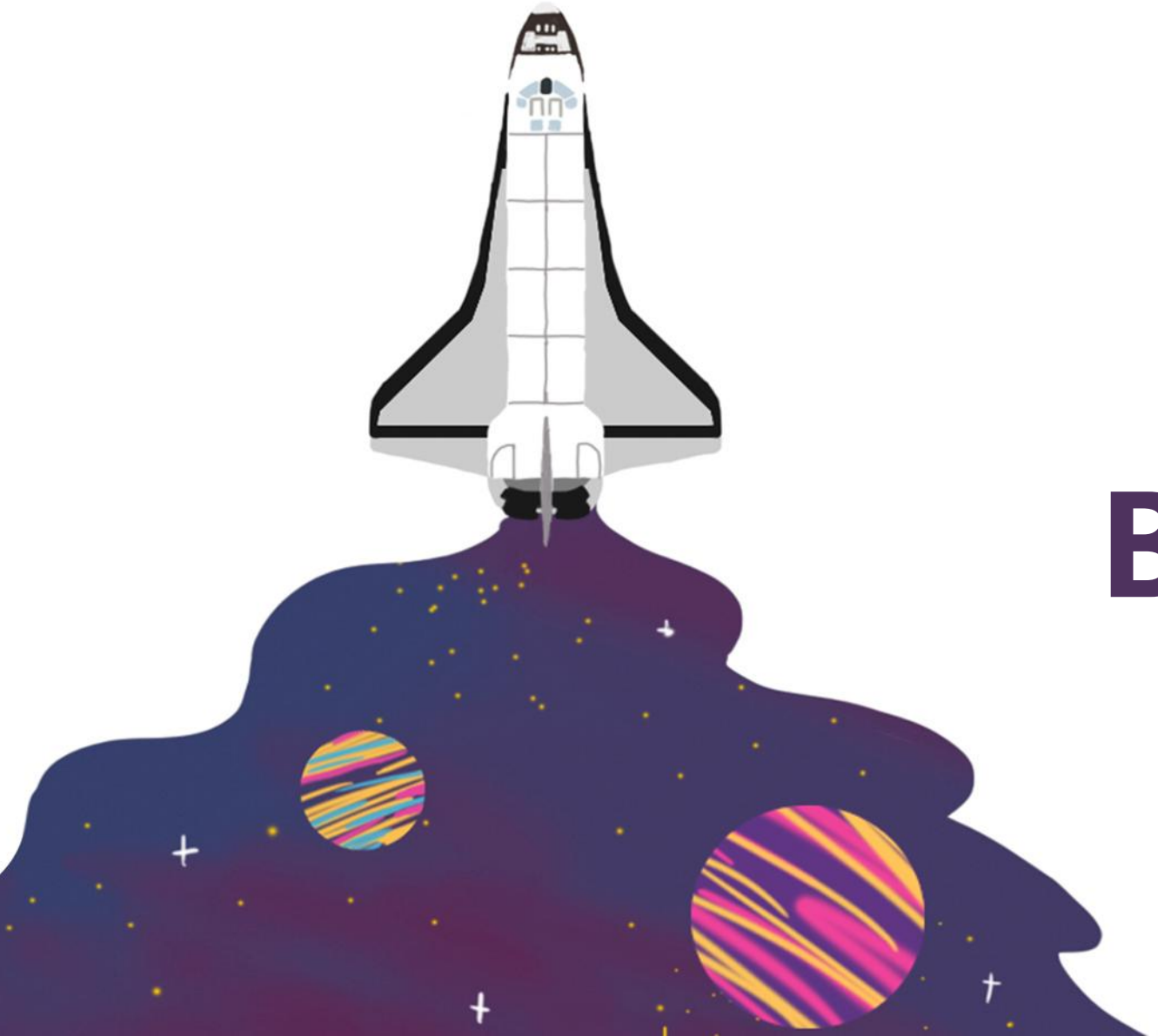

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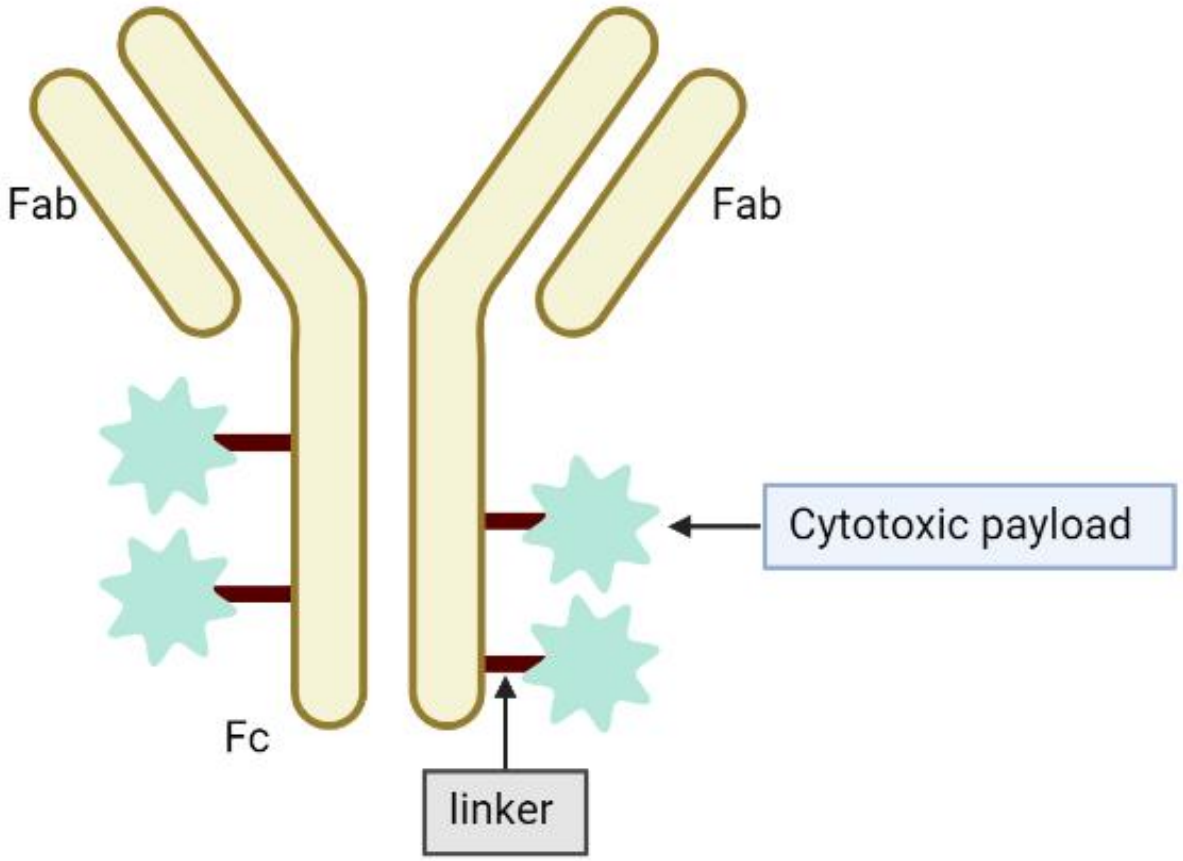
Background

1 Background



Antibody Drug Conjugate (ADC)

Cytotoxin + **Antibody**

↓ ↓

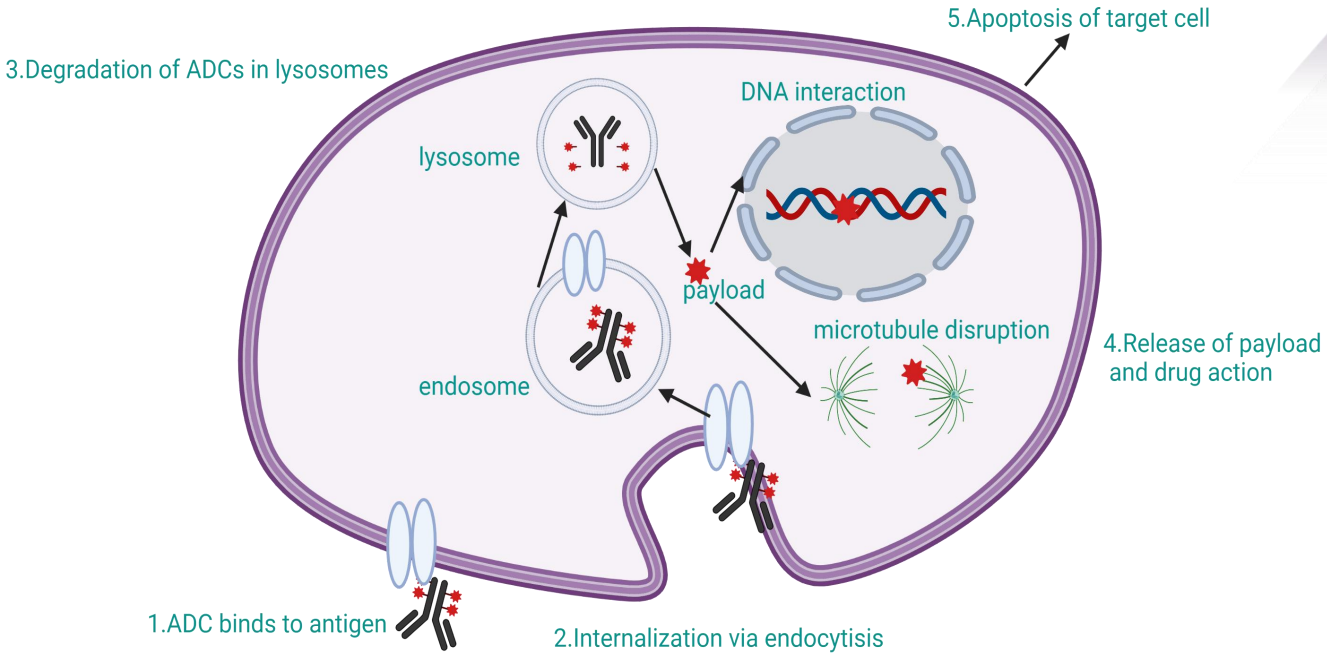
Cytotoxicity **Targeting**

↘ ↙

Tumor treatment

1 Background

ADC Mechanism



Defect

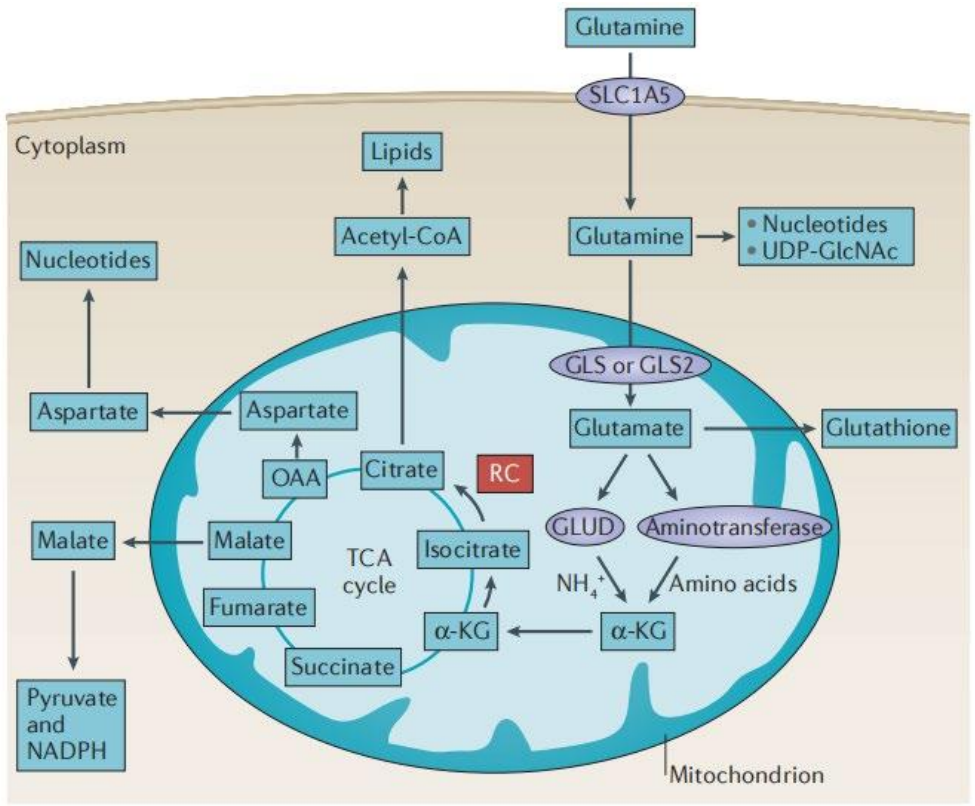
1) The killing effect must be achieved through targeted delivery, endocytosis, and linker breakage. The drug efficiency is low

2) Cytotoxin has high toxicity and side effects, and the "off-target" effect

3) The therapeutic window is small and the dosage is limited

1 Background

Gln metabolism



(Meng M et al, 2010)

Tumor Microenvir-onment

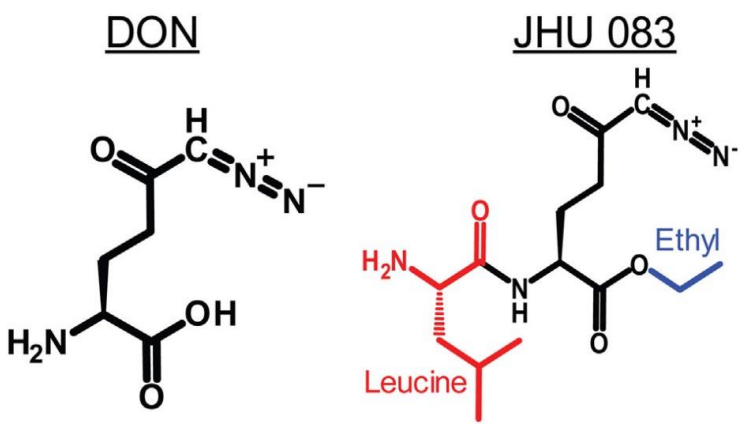
1) Cancer cells upregulate genes involved in glutamine uptake and metabolism, promoting tumor growth

2) Generate a hypoxic, acidic, and nutrient-deficient tumor microenvironment, which is not conducive to the anti-tumor immune response of T cells

1 Background

Glutamine antagonists DON and JHU-083

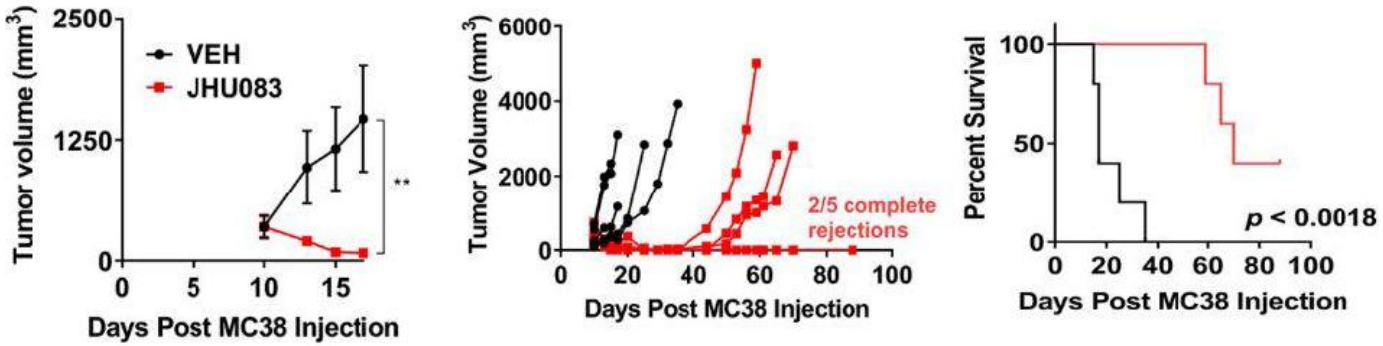
Chemical formula



Block carboxyl and amino groups to prepare prodrugs JHU-083, fixed point release

（Hanaford, A. R. et al，2019）

Efficacy



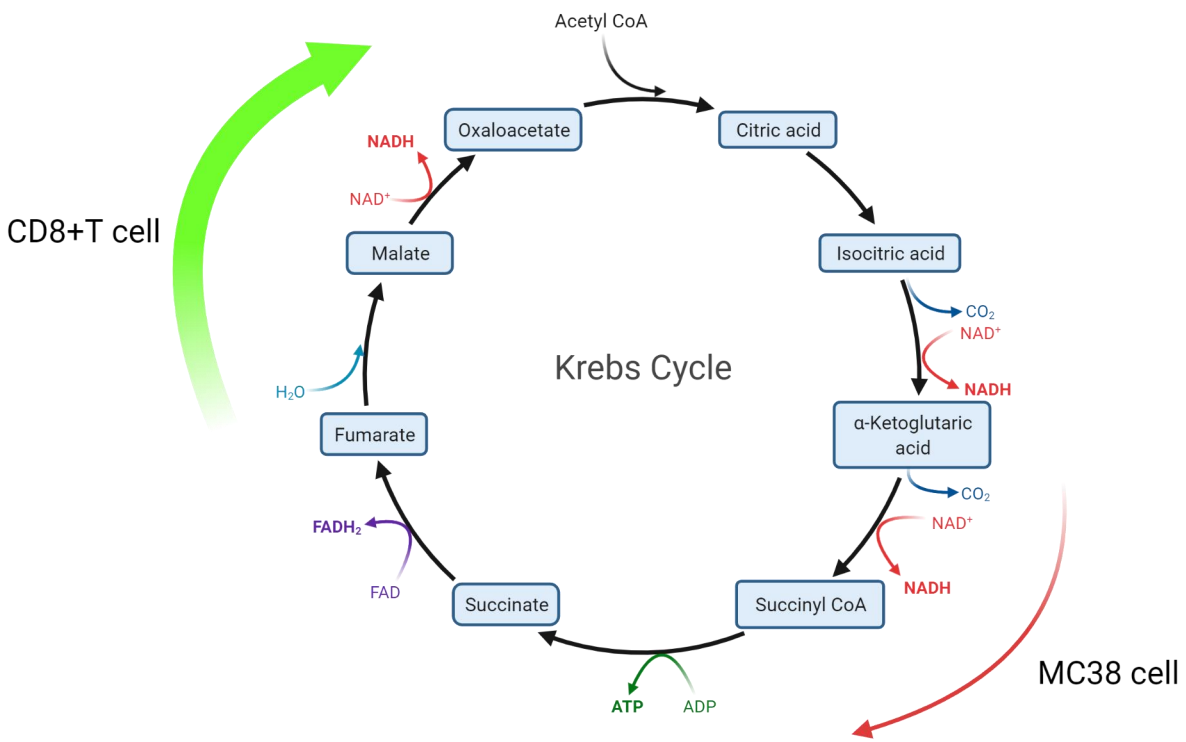
Inhibit the oxidative metabolism and glycolytic metabolism of cancer cells, reverse hypoxia, acidosis and nutrient deficiency in the tumor microenviron-ment, and restore T cell activity

（Leone Robert D et al, 2019）

1 Background

Glutamine antagonists DON and JHU-083

Metabolic differences between T cells and tumor cells



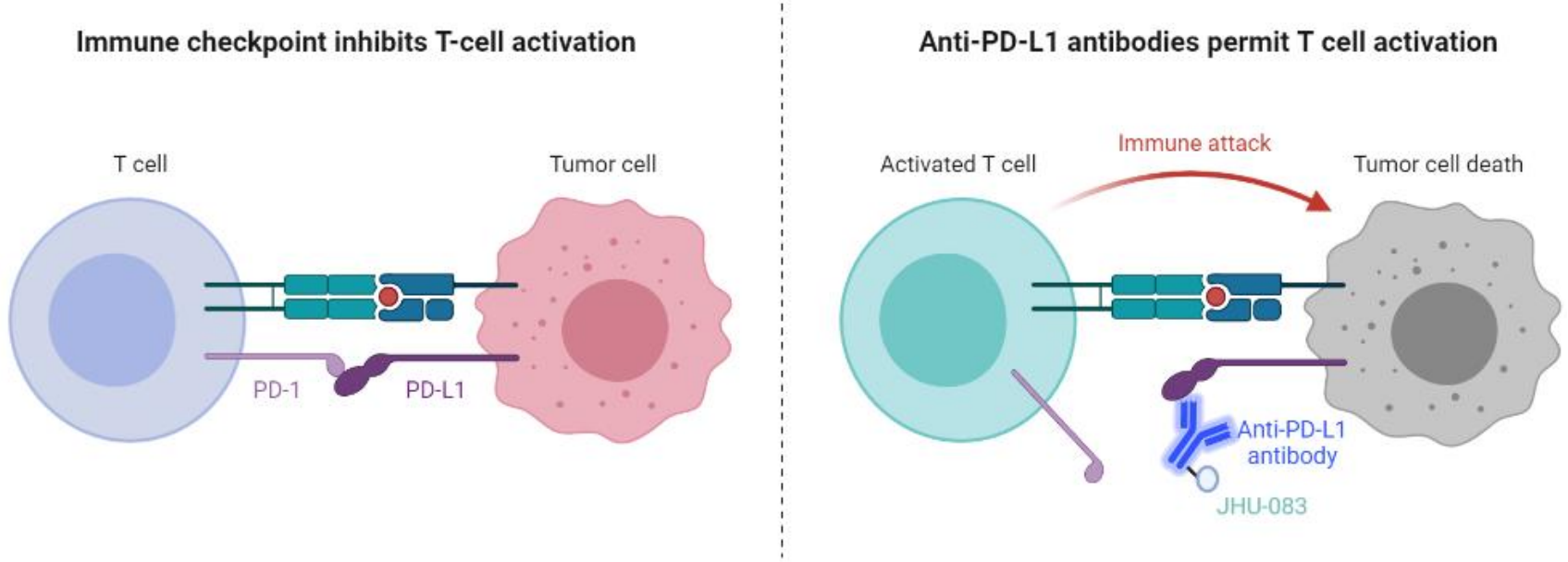
Glutamine antagonists lead to enhanced TCA cycle in CD8+ T cells and attenuated TCA cycle in tumor cell MC38.

Effector T cells respond to glutamine blockade by upregulating oxidative metabolism and exhibiting a long-lived, highly activated phenotype.

1 Background

PD1、PD-L1 and immune escape

Interaction of PD1 and PD-L1



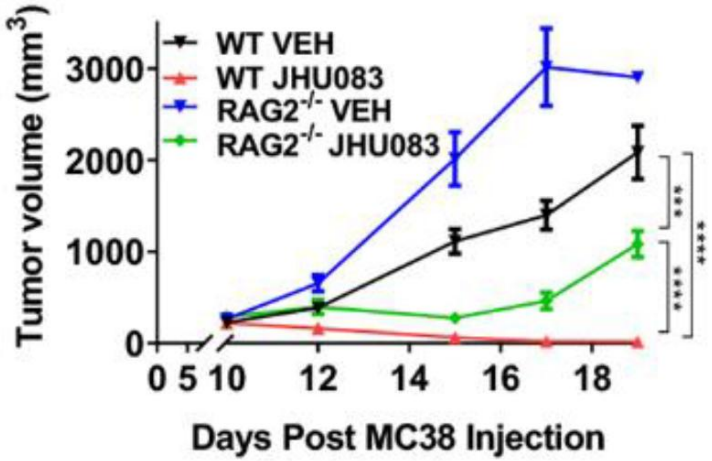
PD-L1 on the surface of tumor cells binds to PD-1 on the surface of T cells, blocking the immune response

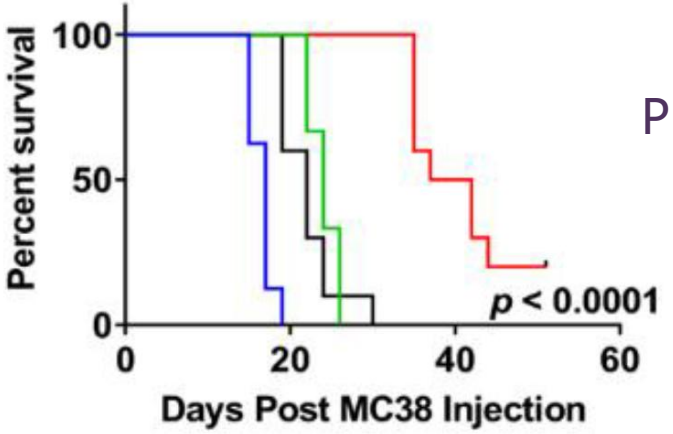
Prepare anti-PD-L1 antibody, block the interaction and restore T cell killing activity

1 研究背景

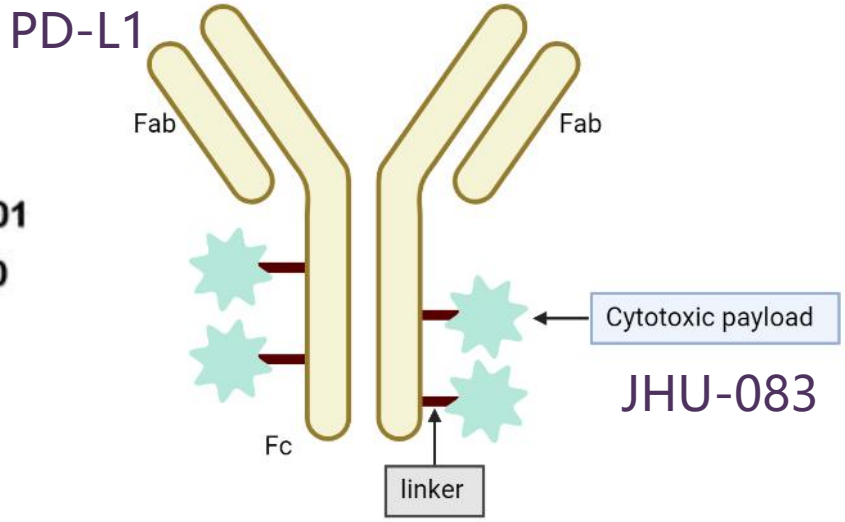
Combination of PD1 mAb with JHU-083

Conjugate JHU-083 to a antibody targeting PD-L1 on tumor cells to achieve specific killing of tumor cells while reducing drug side effects





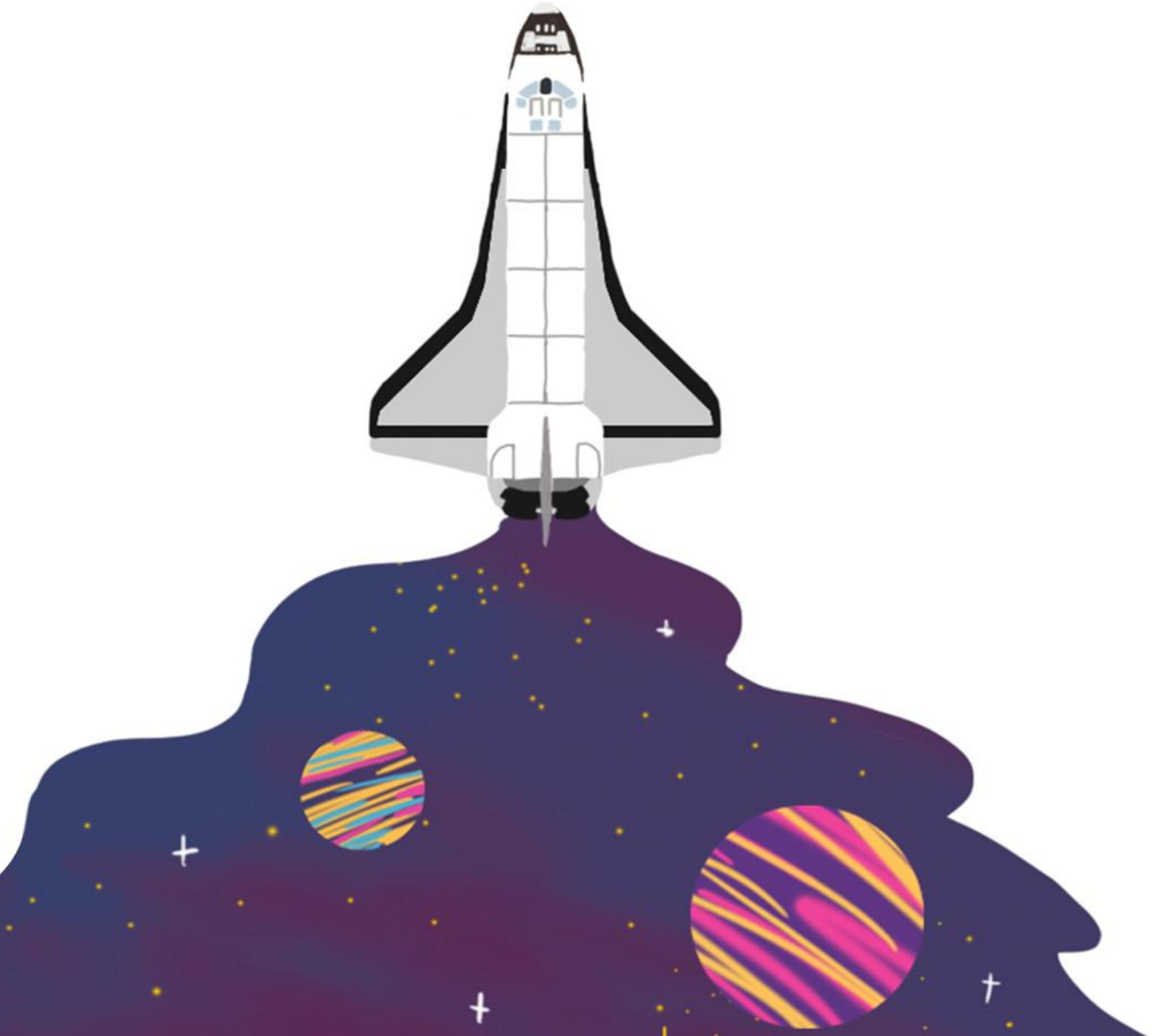
（Leone Robert D et al, 2019）



1) JHU-083 lacks targeting mechanism

2) Combined treatment of tumor cells with JHU-083 and PD1 monoclonal antibody can achieve a multiplied effect

Cleavable acid-sensitive linkers released in the tumor microenvironment



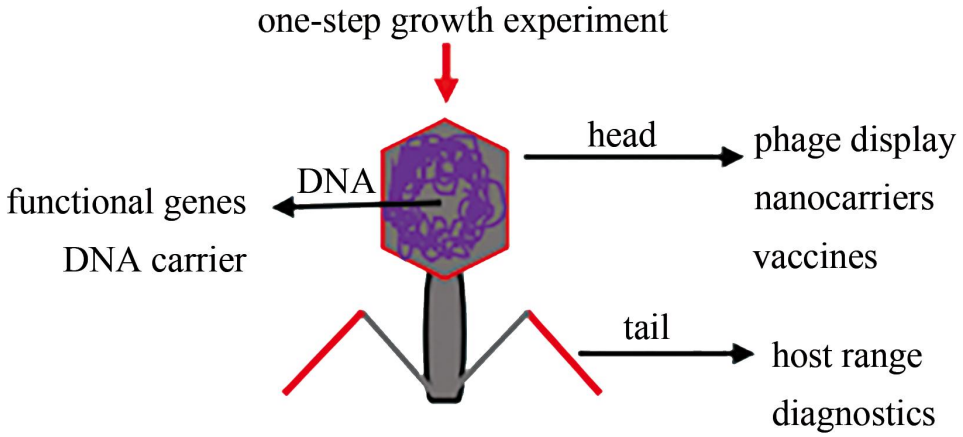
2

Method

2

Antibody Production

Phage display

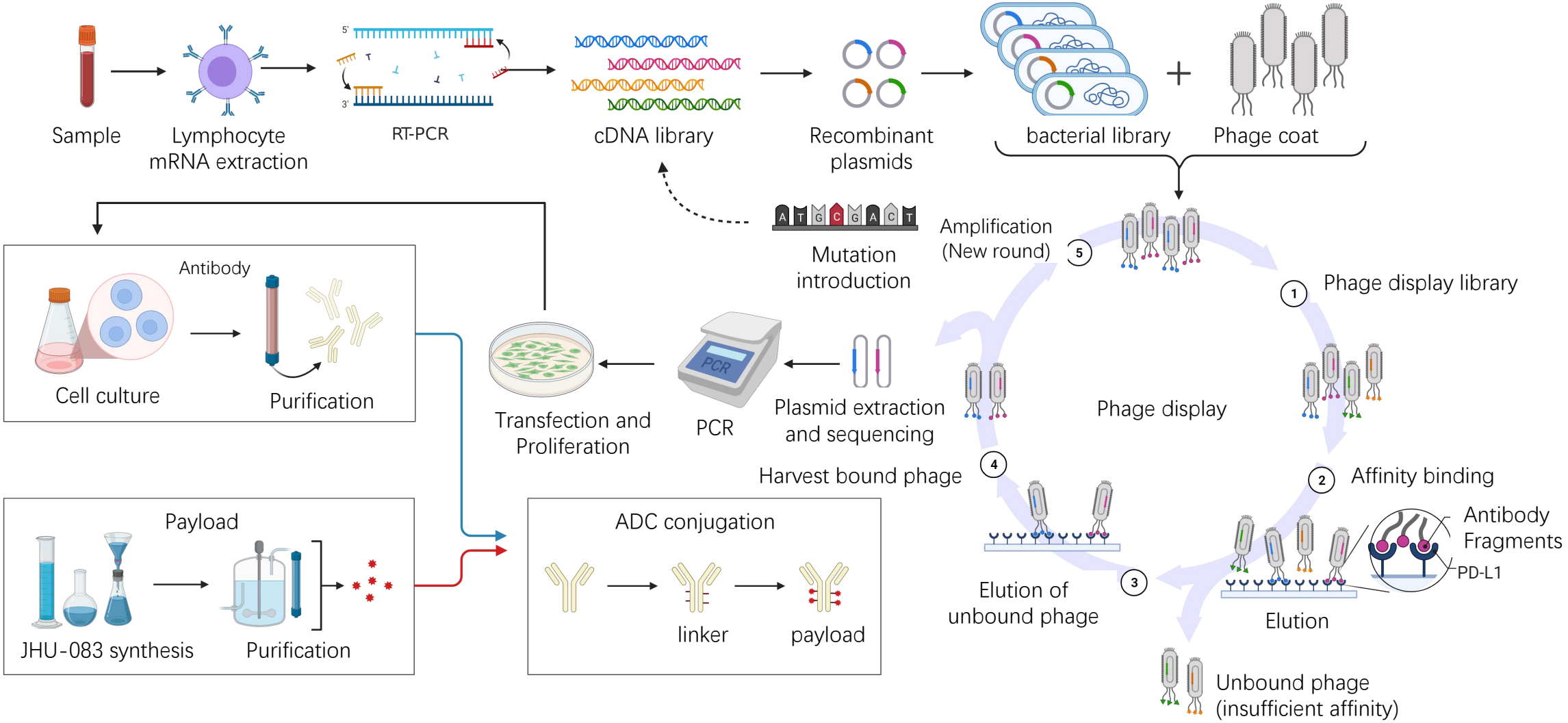


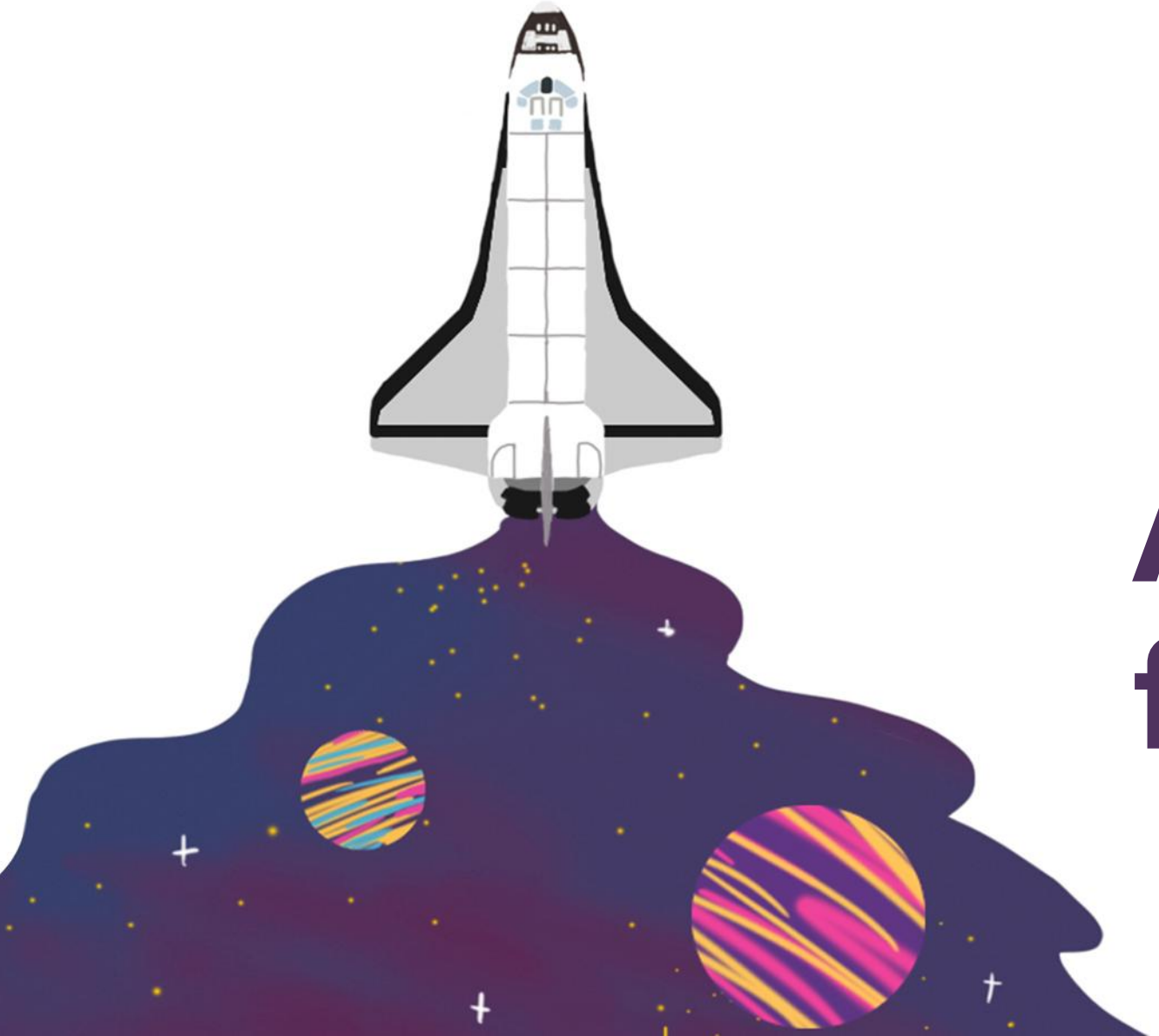
(Yuan S, Ma Y et al. 2020)

Phage display technology can quickly link phenotype selection with genotype. By fusing the polypeptide with the coat protein of the phage, the desired functional polypeptide can be screened out, and the genotype of the corresponding polypeptide can be easily obtained by sequencing.

2 Antibody Production

Phage display





3

Antibody function test

3

Antibody function test

Experimental Design Ideas

General idea

I

Antibody properties

Antibody aggregation, Kd, Ec50, etc.

II

1+1>1?

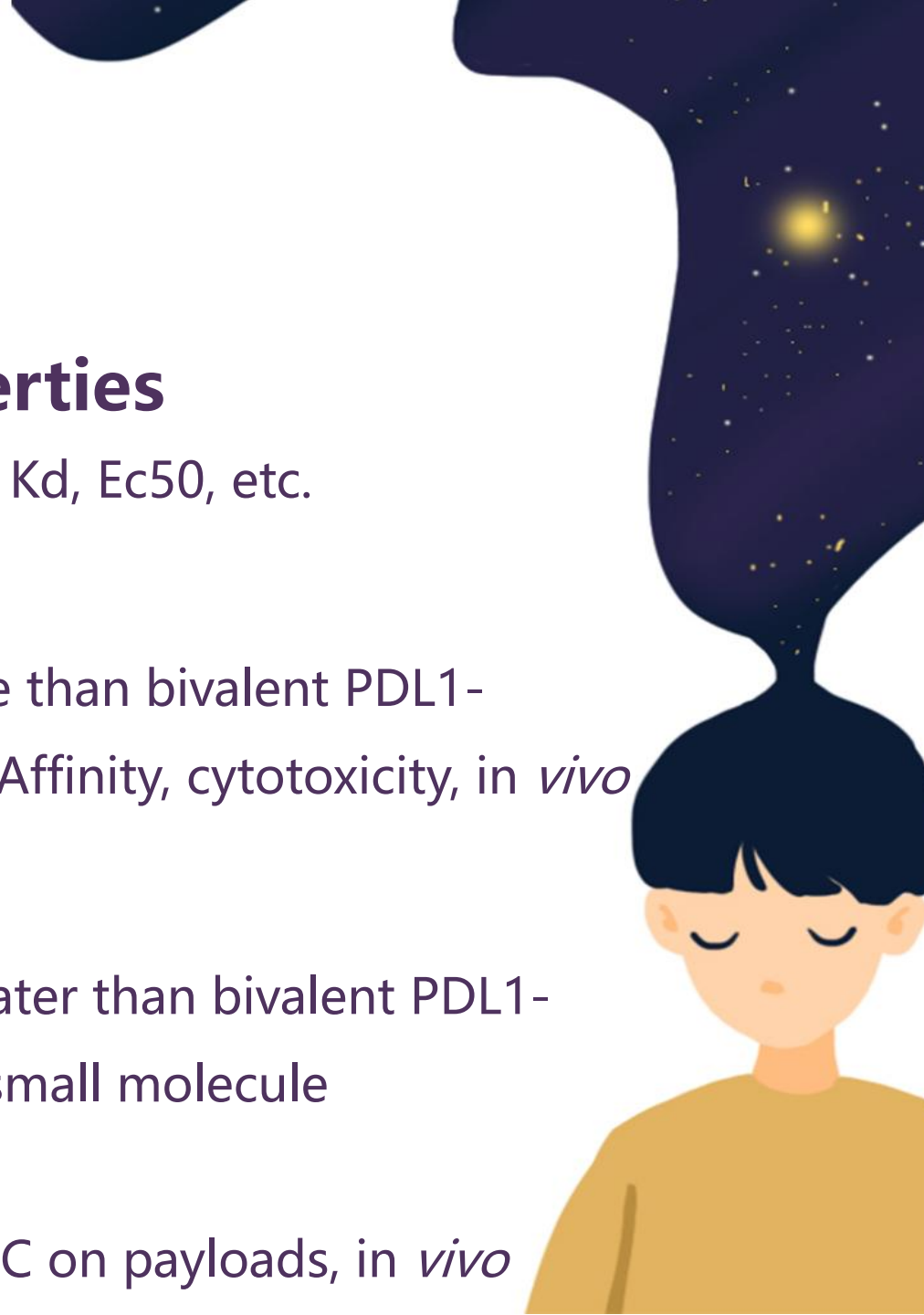
Are ADCs more effective than bivalent PDL1-antibodies or payloads?Affinity, cytotoxicity, in *vivo*

III

1+1>2?

Is the effect of ADC greater than bivalent PDL1-antibody and JHU-083 small molecule combination therapy?

Enrichment effect of ADC on payloads, in *vivo*



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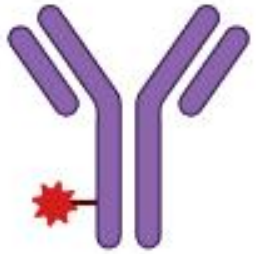
Antibody function test

Experimental Design Ideas

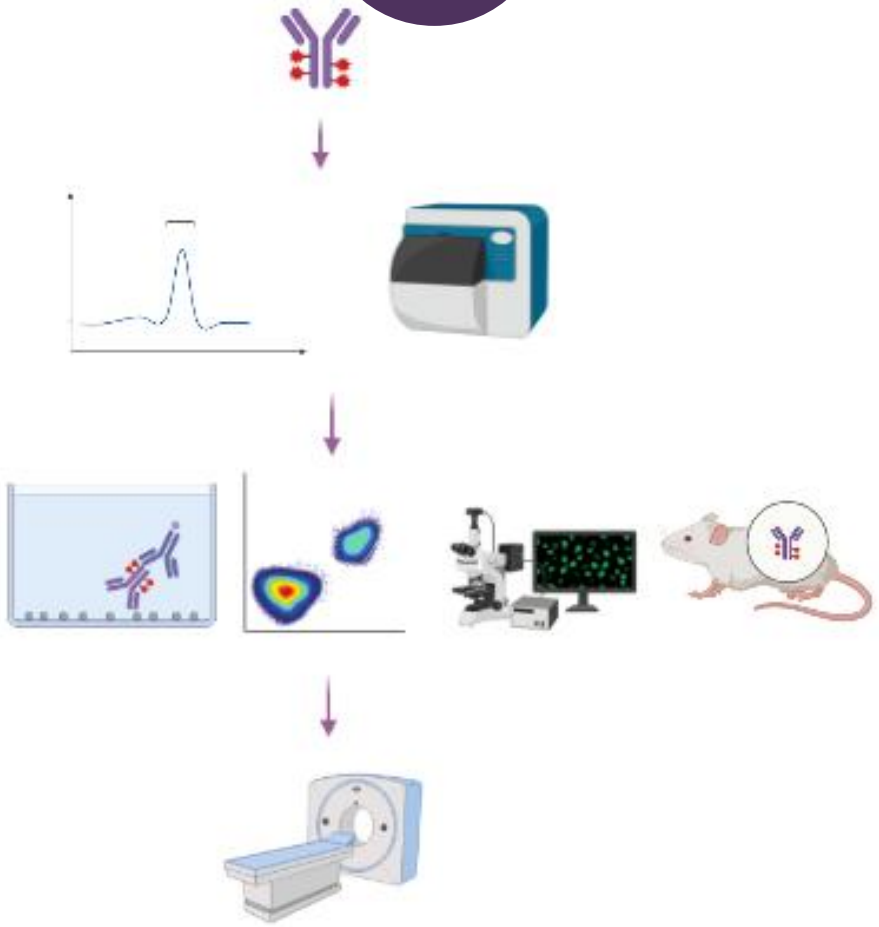
Construct three antibodies

α-PD-L1



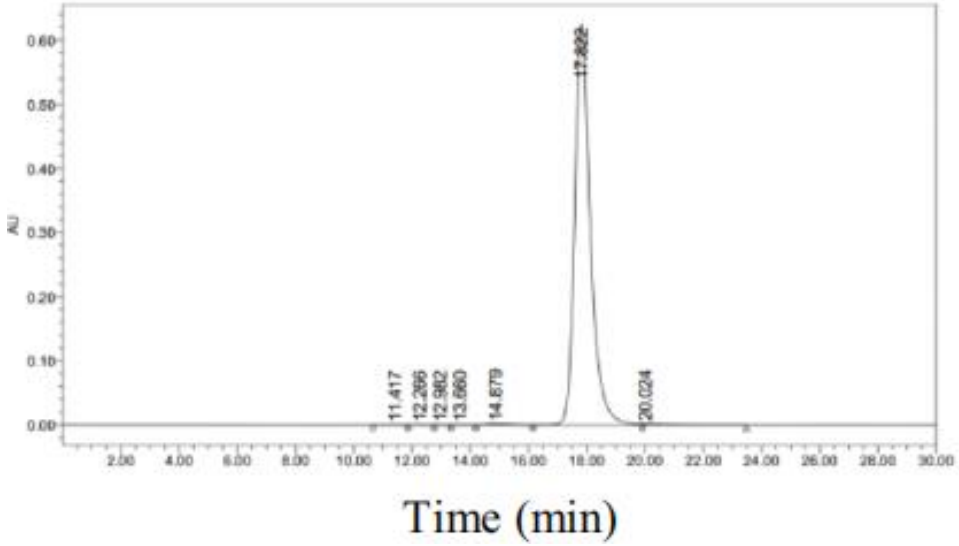
ADC

Control hIgG



Antibody properties

Aggregation-Gel Chromatography



There should only be a single peak if the antibody does not aggregate

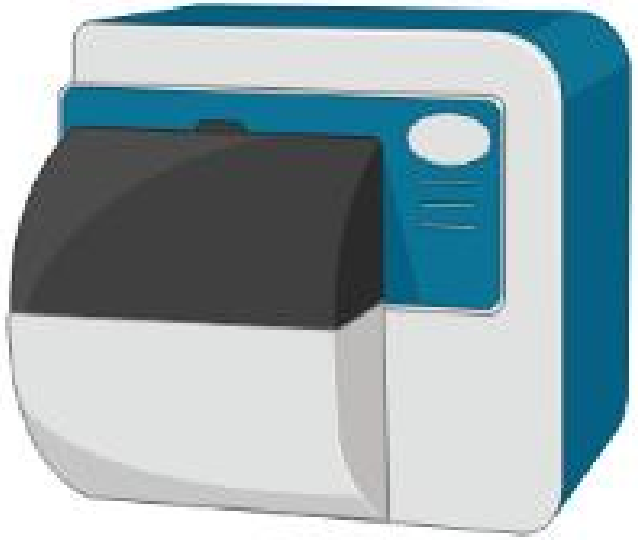
（Wang Y, Ni H, Zhou S et al，2021）

Condition：
50mL PBS+0.1% BSA+0.05% Tween-20 buffer

Improvement:
In the case of multi-peak, change the pH and ionic strength of the solution until the chromatography result is a single peak

Antibody properties

Kd-biofilm interferometry（BLI）

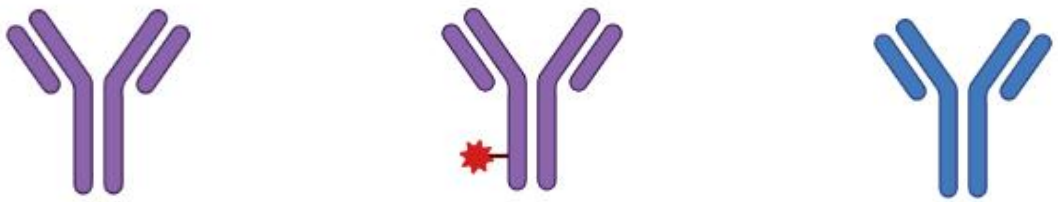


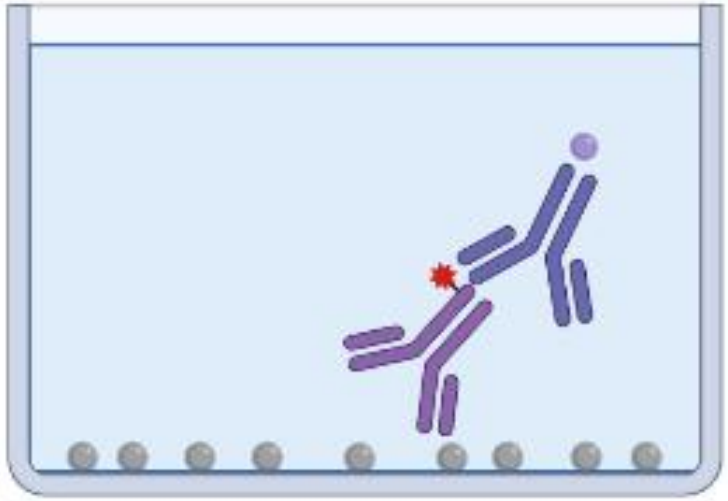
Method:
CHO-S overexpression human PD-L1 cell line (CHO-hPD-L1) was constructed with CHO-S Kit. Direct incubation of CHO-hPD-L1 cells with ADC

Improvement:
If the K_d value is too small, select antibodies with higher affinity from the antibody library

II Antibody Effect

Affinity-ELISA



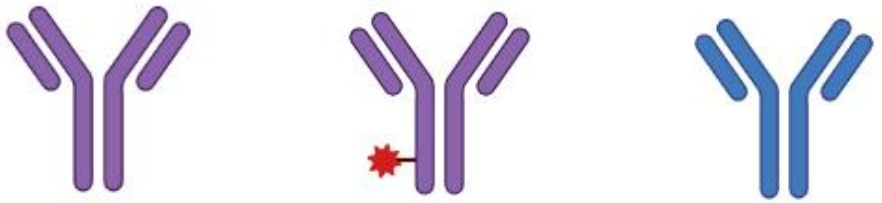


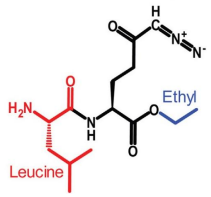
3 GROUPS：
α-PD-L1、ADC、hIgG

Method:
Construct CHO-hPD-L1 cell line, block;
Add 1st antibody; wash 3 times;
Add 2nd antibody PE-anti-human IgG and incubate for 30 minutes;
Calculation of MFI values by flow cytometry

II Antibody Effect

Cytotoxicity test-PI staining





4 GROUPS：

α-PD-L1、ADC、hIgG、JHU-083

Training conditions:

Peripheral blood mononuclear cells (PBMCs), tumor cells

Drugs under concentration gradient (0.0001 ~ 10,000 ng/mL)

Detection:

Propidium iodide (PI) staining;

Fluorescence intensity was detected by flow cytometry;

Draw the cytotoxicity curve according to the fluorescence intensity to get EC50

II Antibody Effect

T Cell Activation Level - Mixed Lymphoid Reaction (MLR)

4 GROUPS：
α-PD-L1、ADC、hIgG、JHU-083

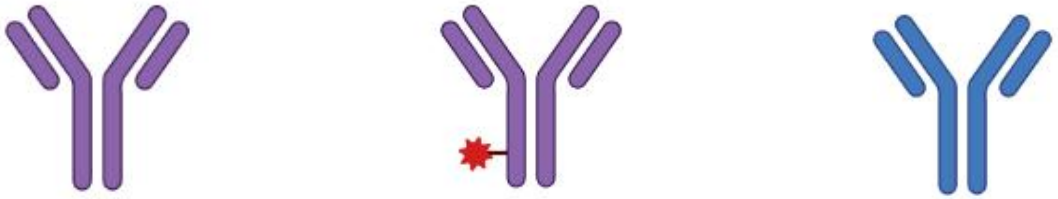
Training conditions:
Peripheral blood mononuclear cells (PBMCs) at the optimal E:T ratio,
tumor cells,
Drugs under concentration gradient (0.0001 ~ 10,000 ng/mL)

Detection:
Changes in the number of T cells:
anti-CD3-FITC, anti-CD25-APC, anti-CD69-PE staining, % change of flow cytometry T cells, fluorescence intensity of T cell activation markers.
Cytokine content:
ELISA Kit for quantitative determination of IL-2, IL-6, IFN-γ and TNF-α.

II Antibody Effect

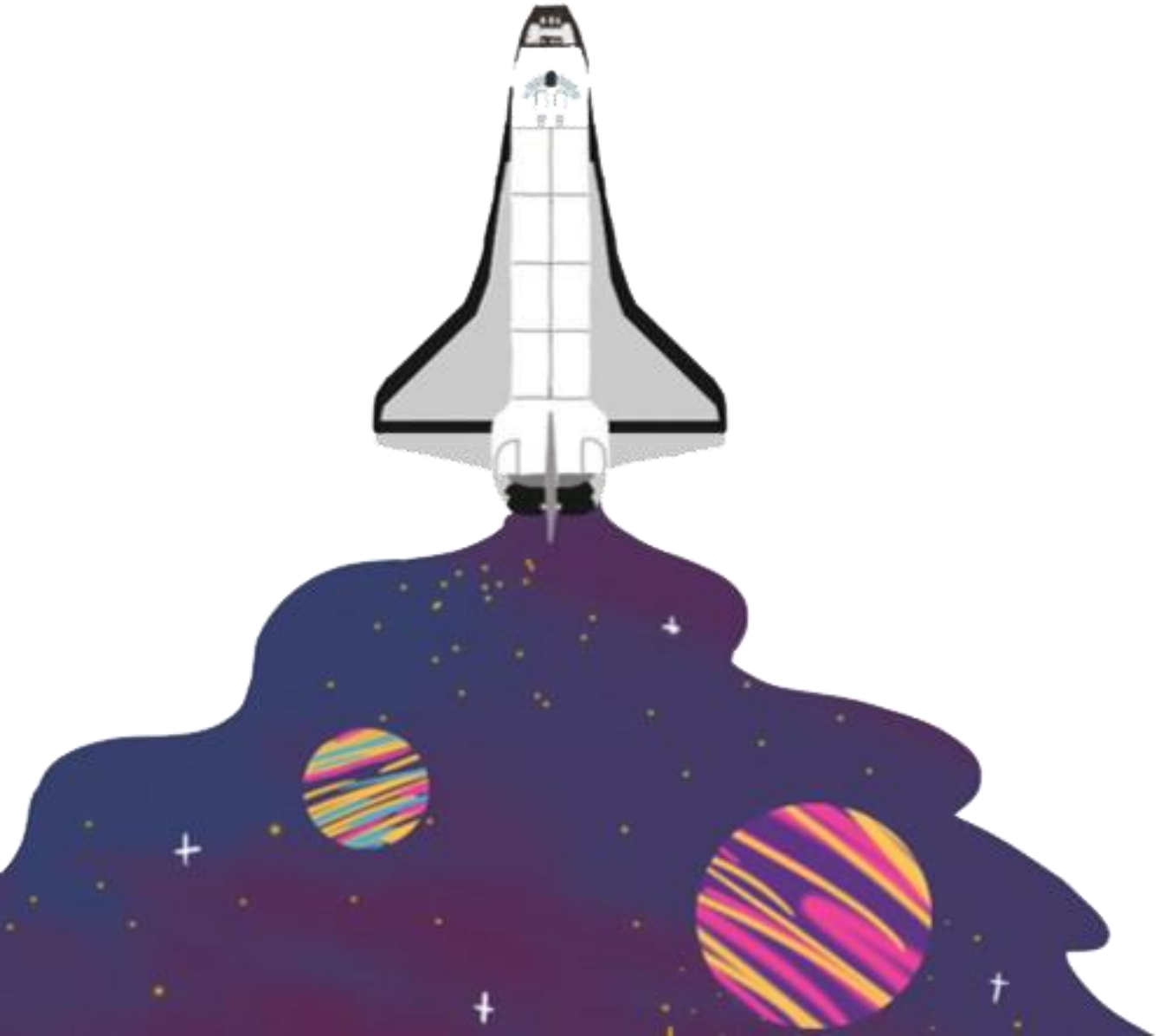
Macrophage Phagocytosis - Imaging with Fluorescent Labeling

3 GROUPS：
α-PD-L1、ADC、hIgG



Experimental conditions:
ADC-treated tumor cells were labeled with red fluorescence (CMPTX);
α-PD-L1-treated tumor cells were labeled with green fluorescence (CMFDA);
hIgG-treated tumor cells are labeled with blue fluorescence (CMF2HC);
Incubate with the same concentration of macrophages for 24 hours

Detection:
Photographs were taken every half hour, and the remaining tumor cells were counted.



4

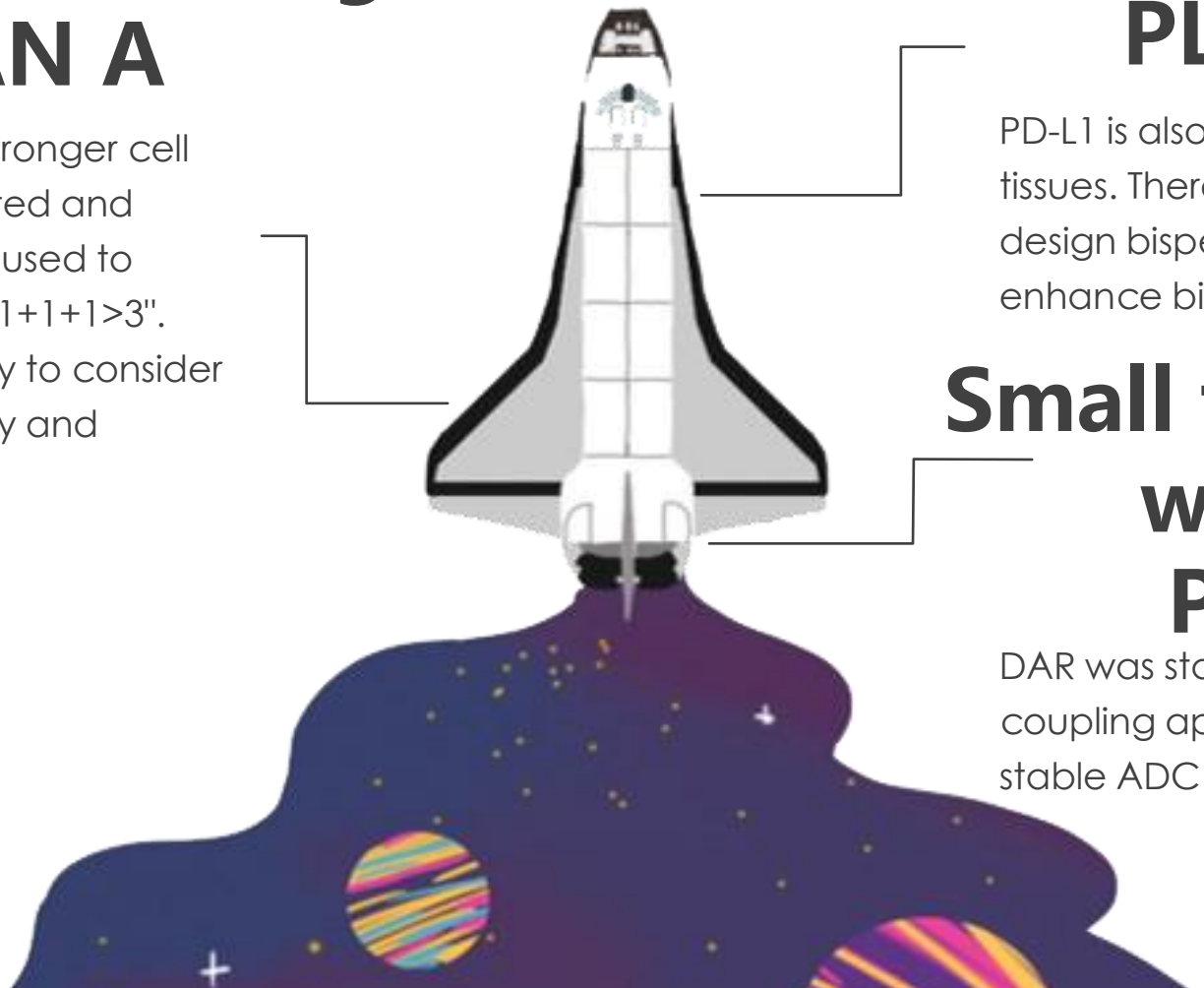
Summary

4 Summary

Possible improvements

Not cytotoxic enough? PLAN A

Small molecules with stronger cell killing power are selected and multivalent coupling is used to achieve the effect of "1+1+1>3". However, it is necessary to consider improving linker stability and endocytic efficiency.



Lack of specificity? PLAN B

PD-L1 is also expressed in some normal tissues. There is therefore an option to design bispecific antibodies to enhance binding specificity.

Small therapeutic window? PLAN C

DAR was stabilized using a site-directed coupling approach. A more uniform and stable ADC conjugate is obtained.